

**MINOR PROJECT REPORT**  
**Design and Development of a Full Stack**  
**Task Management Web Application**

TaskFlow

**Submitted by**

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Domain: Full Stack Development

Project Type: Individual Minor Project

Project Duration: 7 Days

Quillance | Full Stack Development

## 1. Project Overview

TaskFlow is a full-stack task management web application developed as an Individual Minor Project in Full Stack Development. The application allows users to register and log in, create and manage tasks, update task details, track task status, set priorities and due dates, and monitor progress from a dashboard.

## 2. Project Objective

The objective is to demonstrate how the frontend, backend, REST APIs, authentication, database, and application logic work together in a complete full-stack web application.

Application Flow: User Interface → Frontend → API → Backend → Database → Response → Updated Interface

## 3. Problem Statement

Managing daily tasks through unstructured notes or separate tools can make it difficult to track pending work, completed work, priorities, and due dates. TaskFlow provides a centralized web interface for storing and managing tasks using a database-backed application.

## 4. Scope of the Project

- User registration and login.
- Authenticated task management.
- Create, read, update, and delete operations for tasks.
- Pending and Completed task status.
- Task title, description, due date, priority, and category.
- Dashboard counters and completion progress.
- Search and filtering by status and priority.
- Responsive interface for desktop, tablet, and mobile screens.

## 5. Functional Features

### 5.1 Authentication

- User registration and login.
- JWT-based authentication.
- Password hashing using bcryptjs.
- Protected task APIs.
- Logout functionality.

### 5.2 Task Management

- Create, view, edit, and delete tasks.
- Mark tasks as Pending or Completed.
- Set title, description, due date, priority, and category.
- Delete confirmation.

### 5.3 Dashboard

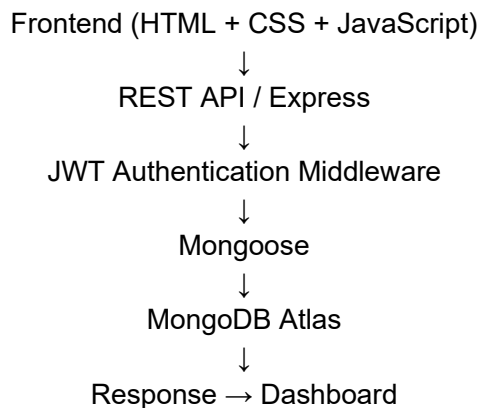
- Total, Pending, Completed, and High Priority counters.
- Task completion progress.
- Search tasks.
- Filter by status and priority.
- Responsive task cards.

## 6. Technology Stack

| Layer               | Technology                                      |
|---------------------|-------------------------------------------------|
| Frontend            | HTML5, CSS3, JavaScript, Bootstrap 5            |
| Backend             | Node.js, Express.js                             |
| Database            | MongoDB Atlas                                   |
| ODM / Security      | Mongoose, JWT, bcryptjs                         |
| Other Backend Tools | CORS, dotenv                                    |
| Development Tools   | Visual Studio Code, Thunder Client, Git, GitHub |

## 7. System Architecture

The application follows a simple full-stack architecture in which the frontend communicates with REST APIs exposed by Express. JWT authentication middleware protects task operations. Mongoose handles communication with MongoDB Atlas.



## 8. Authentication Flow

During registration, user information is stored in MongoDB with a hashed password. During login, credentials are verified and a JWT token is generated. Protected task requests send the token using the Authorization header, and the authentication middleware identifies the logged-in user.

Register → MongoDB → Login → Credentials Verified → JWT Token  
→ Protected Task APIs → Authentication Middleware → User-specific Tasks

## 9. CRUD Operations and API

| Operation | Method | Endpoint   | Purpose            |
|-----------|--------|------------|--------------------|
| Create    | POST   | /api/tasks | Add a new task     |
| Read      | GET    | /api/tasks | Retrieve logged-in |

|        |        |                |                               |
|--------|--------|----------------|-------------------------------|
|        |        |                | user's tasks                  |
| Update | PUT    | /api/tasks/:id | Update task details or status |
| Delete | DELETE | /api/tasks/:id | Delete a task                 |

Authentication endpoints:

| Method | Endpoint           | Description         |
|--------|--------------------|---------------------|
| POST   | /api/auth/register | Register a new user |
| POST   | /api/auth/login    | Authenticate a user |

## 10. Database Design

### 10.1 User Collection

- Name
- Email
- Hashed password
- Created and updated timestamps

### 10.2 Task Collection

- User ID
- Title
- Description
- Due date
- Status
- Priority
- Category
- Created and updated timestamps

Each task is associated with its owner through userId.

## 11. Project Structure

KrushnaManthalkar\_TaskManagement/

```

├── backend/
│   ├── middleware/
│   │   └── authMiddleware.js
│   ├── models/
│   │   ├── User.js
│   │   └── Task.js
│   ├── routes/
│   │   ├── authRoutes.js
│   │   └── taskRoutes.js
│   ├── .env
│   ├── .gitignore
│   ├── package.json
│   ├── package-lock.json
│   └── server.js
├── frontend/
└── index.html

```

```
| | style.css
| | script.js
| | screenshots/
| | README.md
```

## 12. Testing

- User registration
- User login
- JWT authentication
- Task creation
- Task retrieval
- Task editing
- Task deletion
- Pending/Completed status update
- Search
- Status filtering
- Priority filtering
- Dashboard counters
- Progress summary
- Logout
- Re-login and task persistence

API testing was performed using Thunder Client.

## 13. Responsive Design

The application is designed for desktop, tablet, and mobile screens. Authentication pages, dashboard cards, filters, task cards, and action buttons adapt to smaller screens.

## 14. Security

- Passwords are hashed using bcryptjs.
- JWT protects task APIs.
- Users can access only their own tasks.
- Database credentials are stored in environment variables.
- .env is excluded using .gitignore.

## 15. Development Plan

| Day   | Planned Work                                     |
|-------|--------------------------------------------------|
| Day 1 | Project planning, requirements, and UI structure |
| Day 2 | Front-end interface development                  |
| Day 3 | Back-end setup and API development               |
| Day 4 | Database connection and CRUD operations          |
| Day 5 | Authentication and feature integration           |
| Day 6 | Responsive design, validation, and testing       |
| Day 7 | Final testing, documentation, and submission     |

## 16. Screenshots

The project submission includes screenshots demonstrating the login, registration, dashboard, task creation, task list, status update, search/filter, and mobile-responsive interface.

- Login
- Register
- Dashboard
- Add Task
- Task List
- Status Update
- Search & Filter
- Mobile Responsive

## 17. Future Enhancements

- Email task reminders
- Calendar integration
- Advanced analytics
- User profile management
- Drag-and-drop task organization
- Dark/light theme
- Notifications
- Cloud deployment

## 18. Conclusion

TaskFlow demonstrates the complete flow of a database-backed full-stack web application. It combines a responsive frontend, REST APIs, authentication, MongoDB integration, CRUD operations, task tracking, search and filtering, and dashboard-based progress monitoring. The project provides practical experience with frontend development, server-side programming, database integration, API communication, authentication, and testing.

## 19. Project Information

| Item          | Details                                                                |
|---------------|------------------------------------------------------------------------|
| Student       | Krushna Manthalkar                                                     |
| Domain        | Full Stack Development                                                 |
| Project Title | Design and Development of a Full Stack Task Management Web Application |
| Project Name  | TaskFlow                                                               |
| Project Type  | Individual Minor Project                                               |
| Duration      | 7 Days                                                                 |